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(54) **WAVEGUIDE APPARATUS WITH HIGH SPEED DUAL CHANNEL WIRELESS CONTACTLESS ROTARY JOINT**

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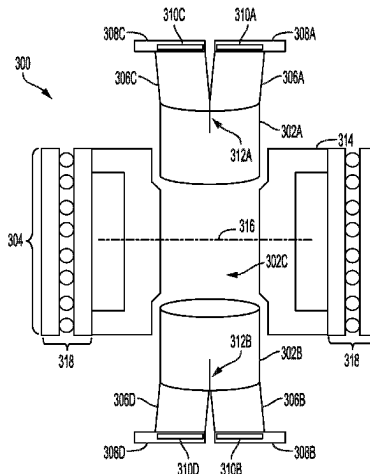
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(57) **ABSTRACT**

A vehicle having a communication system is disclosed. The system includes two electrical couplings, coupled by way of a rotary joint having a bearing waveguide. Each electrical coupling includes an interface waveguide configured to couple to external signals. Each electrical coupling also includes a waveguide section configured to propagate electromagnetic signals between the interface waveguide and the bearing waveguide of the rotary joint. Additionally, the rotary joint is configured to allow one electrical coupling to rotate with respect to the other electrical coupling. An axis of rotation of the rotary joint is defined by a center of a portion of the waveguides. Yet further, the rotary joint allows

(Continued)



electromagnetic energy to propagate between the wave-guides of the electrical couplings.

20 Claims, 7 Drawing Sheets

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continuation of application No. 16/533,519, filed on Aug. 6, 2019, now Pat. No. 11,152,675, which is a continuation-in-part of application No. 15/789,533, filed on Oct. 20, 2017, now Pat. No. 10,522,887.

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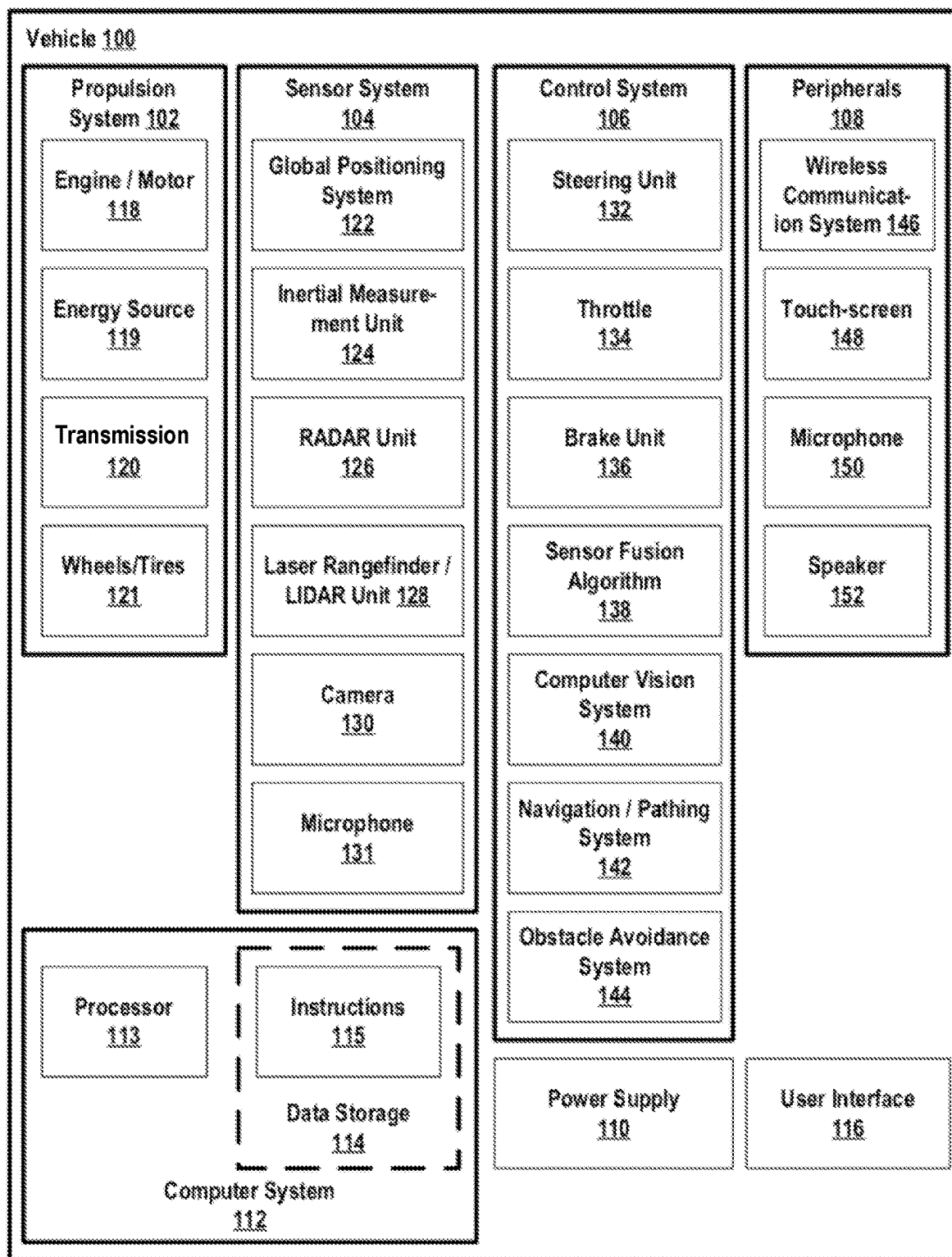


Fig. 1

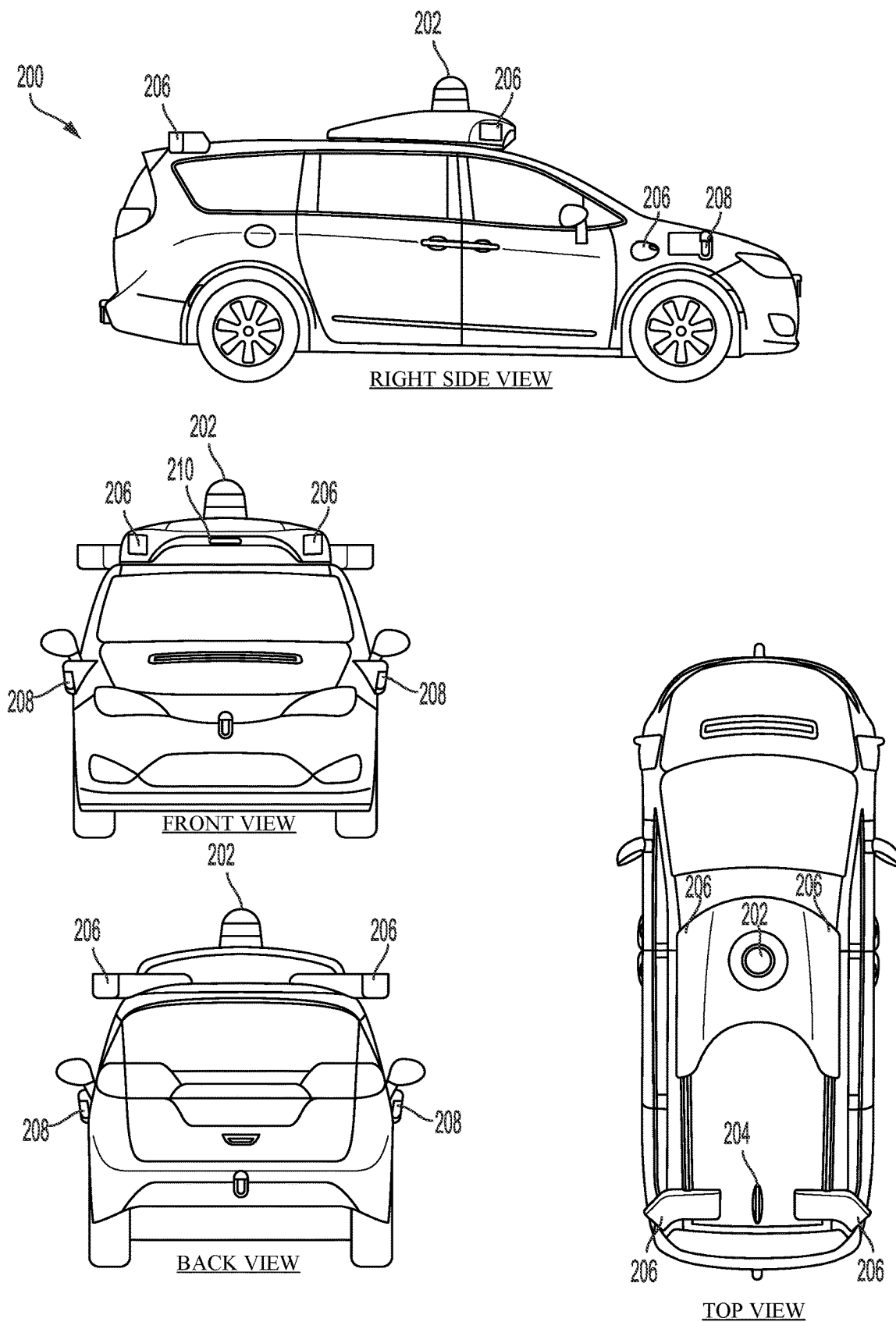


Fig. 2

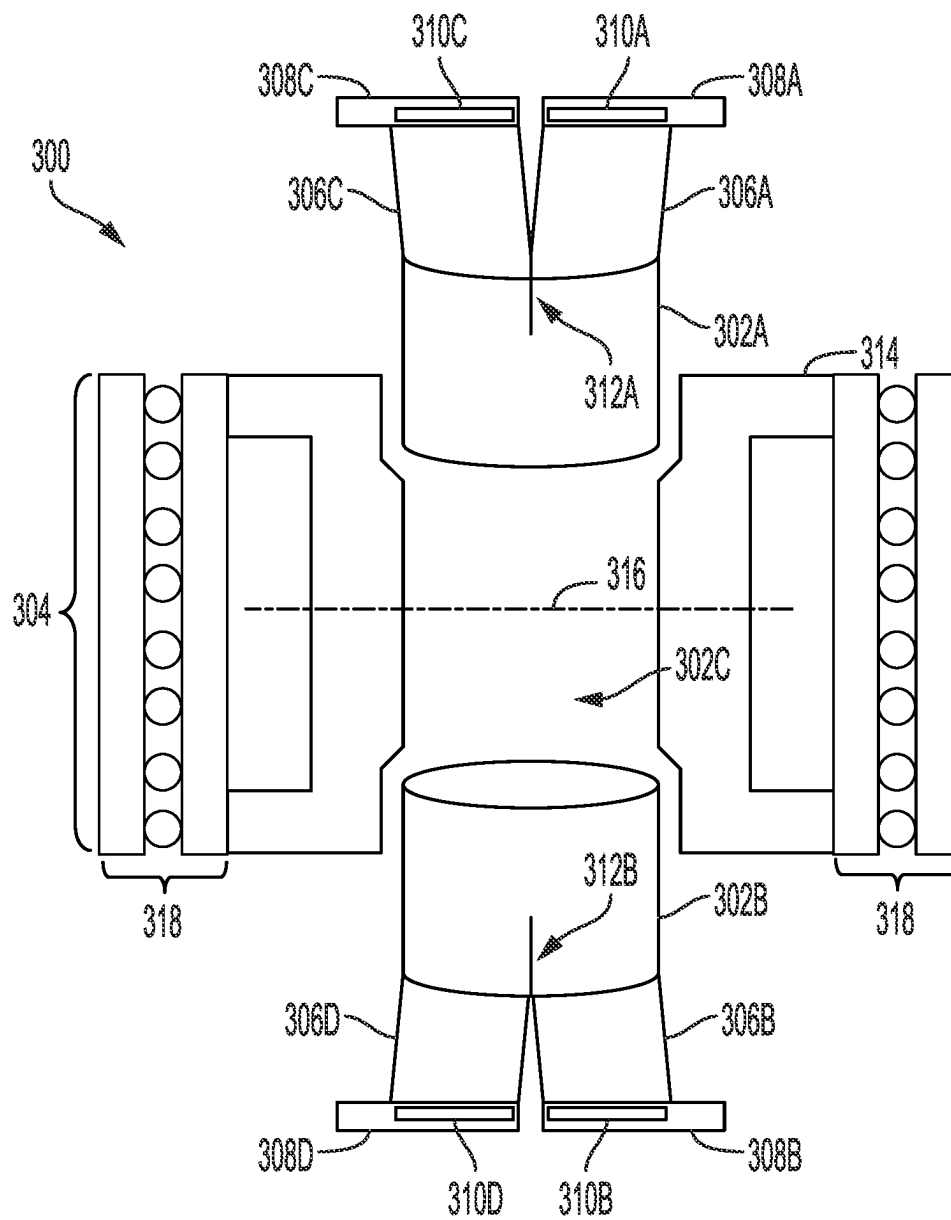


FIG. 3

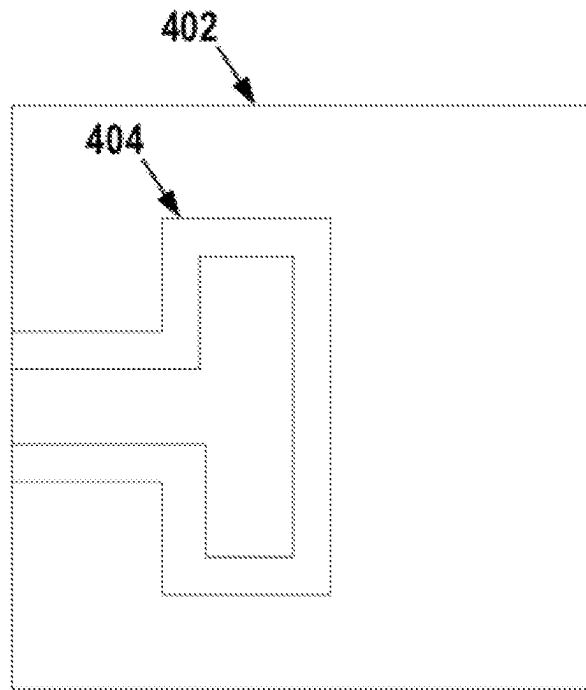


Fig. 4A

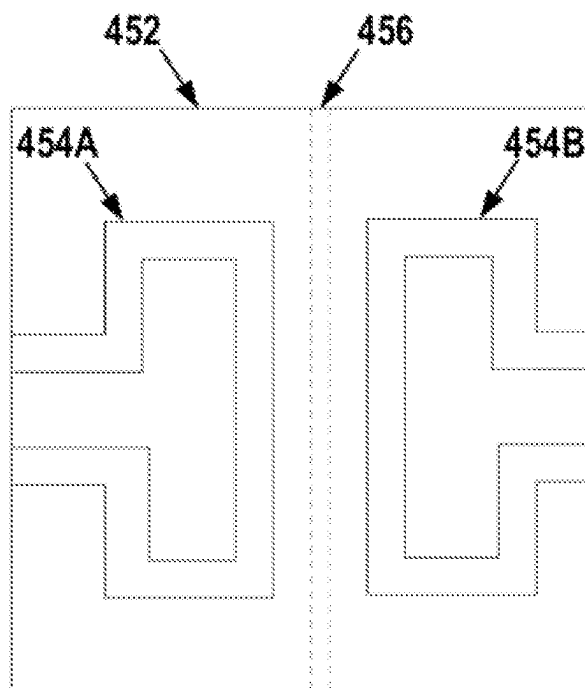


Fig. 4B

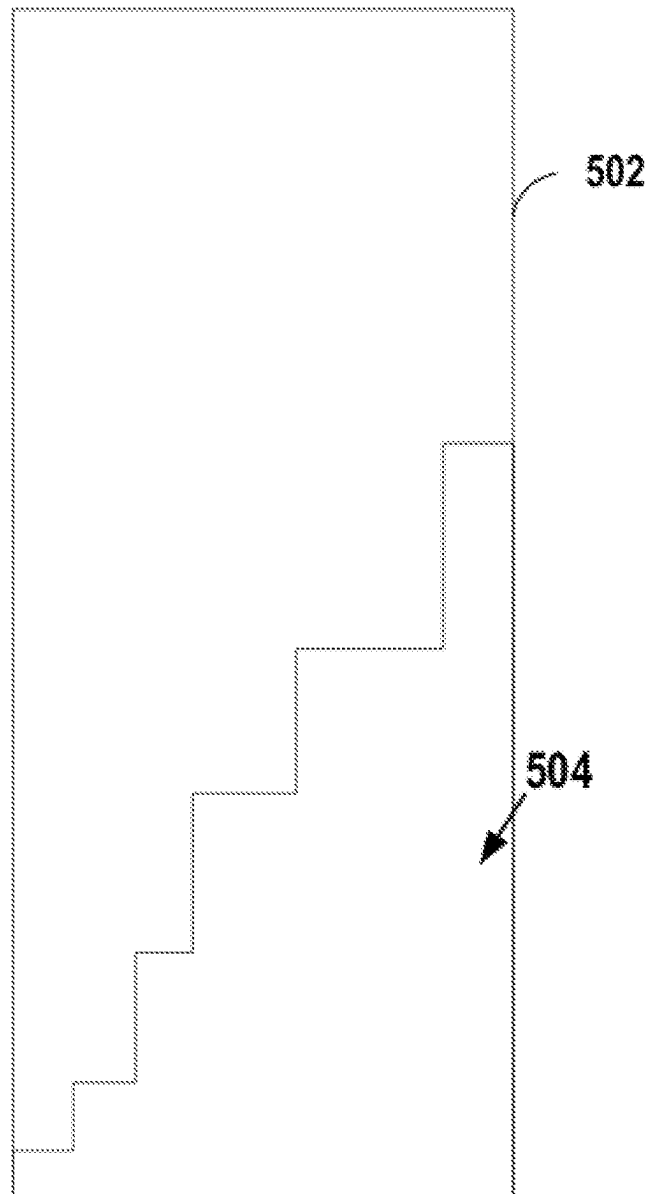


FIG. 5

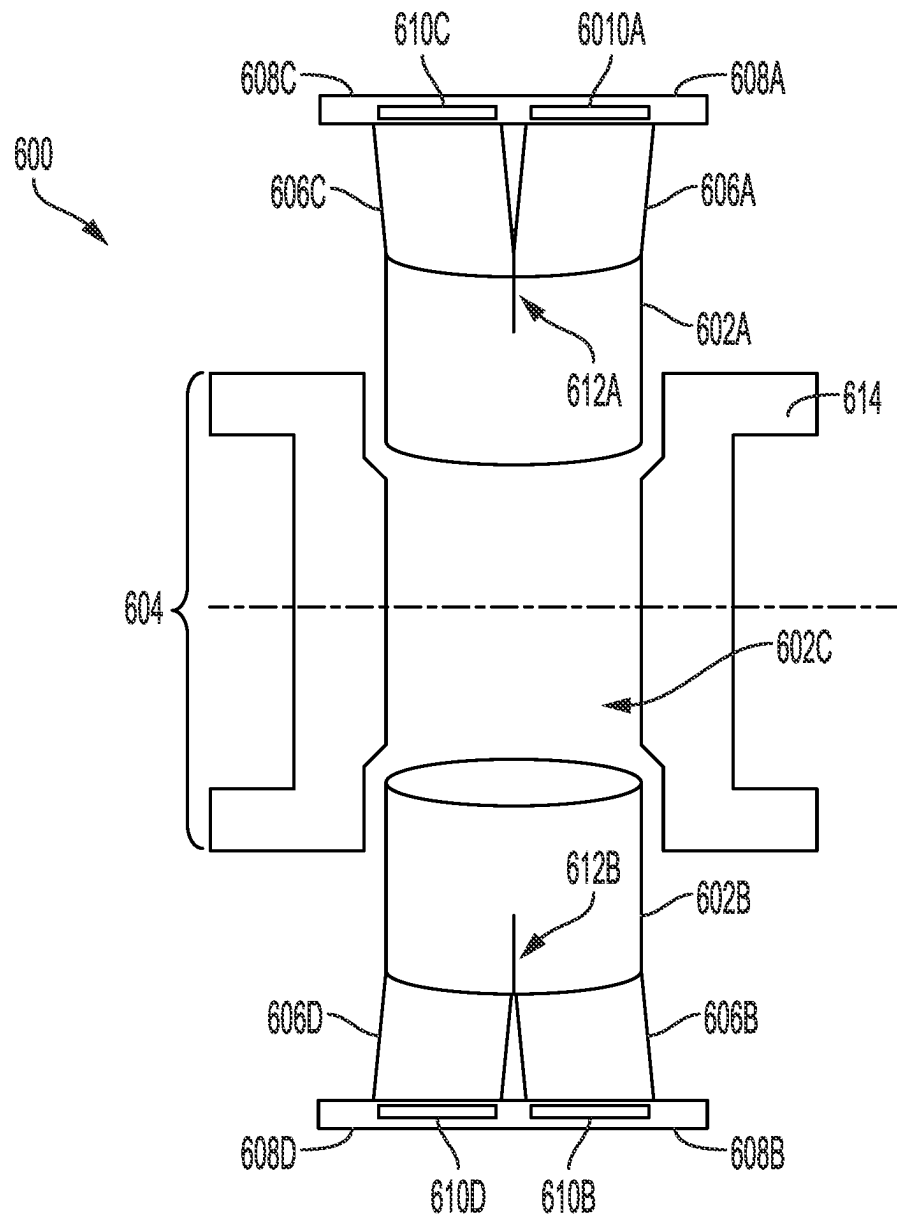
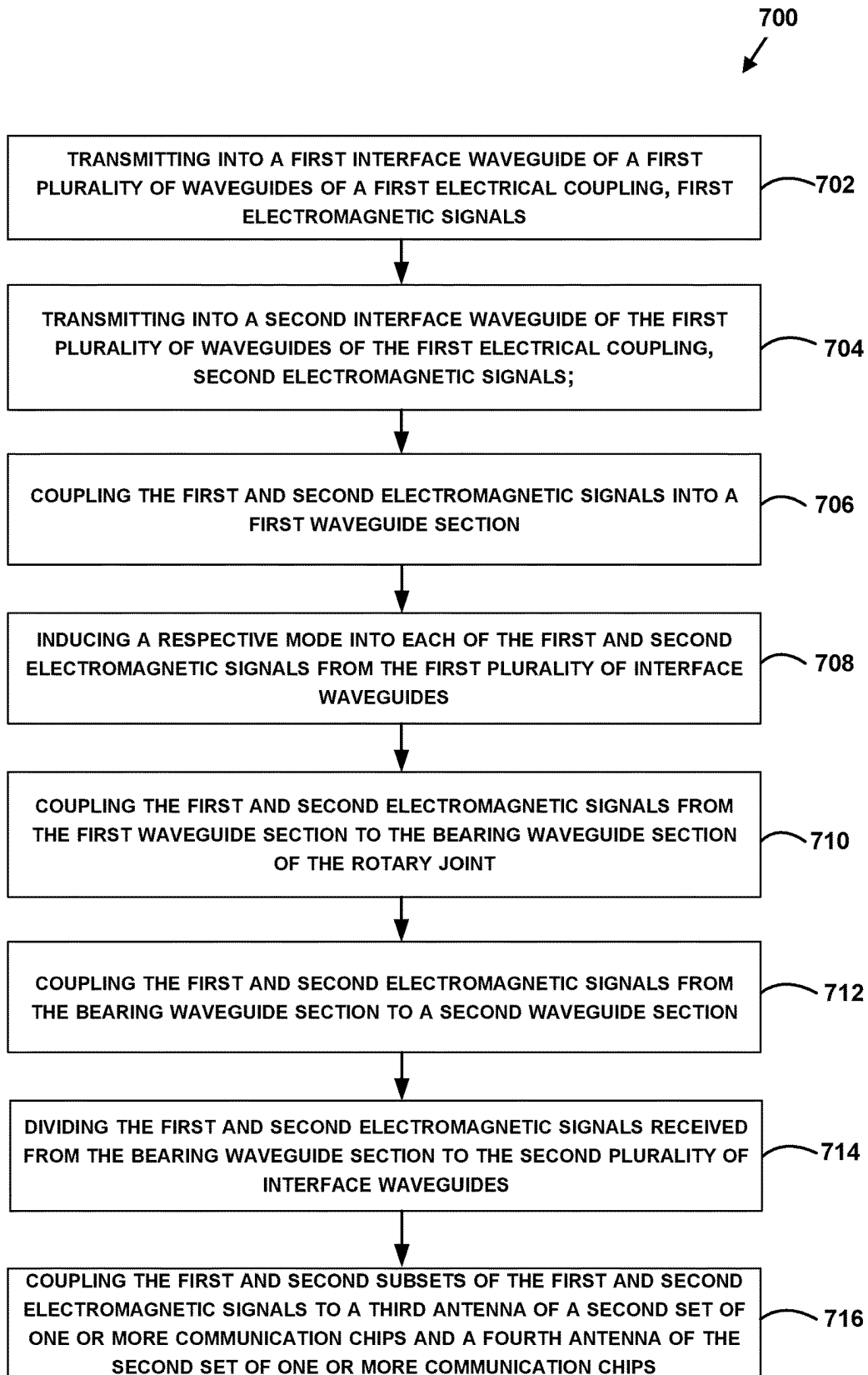


FIG. 6

**FIG. 7**

WAVEGUIDE APPARATUS WITH HIGH SPEED DUAL CHANNEL WIRELESS CONTACTLESS ROTARY JOINT

CROSS REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/450,098, filed on Oct. 6, 2021, which is a continuation of U.S. patent application Ser. No. 16/533,519 (now U.S. Pat. No. 11,152,675), filed on Aug. 6, 2019 and issued on Oct. 19, 2021, which is a continuation-in-part of and claims priority to U.S. patent application Ser. No. 15/789,533 (now U.S. Pat. No. 10,522,887), filed on Oct. 20, 2017 and issued on Dec. 31, 2019, the entire contents of each is hereby incorporated by reference.

BACKGROUND

Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted as prior art by inclusion in this section.

Vehicles can be configured to operate in an autonomous mode in which the vehicle navigates through an environment with little or no input from a driver. Such autonomous vehicles can include one or more sensors that are configured to detect information about the environment in which the vehicle operates. The vehicle and its associated computer-implemented controller use the detected information to navigate through the environment. For example, if the sensor(s) detect that the vehicle is approaching an obstacle, as determined by the computer-implemented controller, the controller adjusts the vehicle's directional controls to cause the vehicle to navigate around the obstacle.

One such sensor is a light detection and ranging (LIDAR) device. A LIDAR actively estimates distances to environmental features while scanning through a scene to assemble a cloud of point positions indicative of the three-dimensional shape of the environmental scene. Individual points are measured by generating a laser pulse and detecting a returning pulse, if any, reflected from an environmental object, and determining the distance to the reflective object according to the time delay between the emitted pulse and the reception of the reflected pulse. The laser, or set of lasers, can be rapidly and repeatedly scanned across a scene to provide continuous real-time information on distances to reflective objects in the scene. LIDAR, and other sensors, may create large amounts of data. It may be desirable to communicate this data, or a variant of this data, to various systems of the vehicle.

SUMMARY

Disclosed are electrical devices that may be used for the communication of signals to and from the various sensors of the vehicle. For example, one or more sensors may be mounted on the roof of the vehicle, such as in a sensor dome. During the operation of the sensor, the sensor may be rotated, such as by way of being mounted on a rotating platform. Although the sensor and platform are rotating, it may be desirable for the sensor to be in data communication with components on the vehicle, such as a data processor associated with the sensor. Therefore, it may be desirable to have a system to communicate signals between the rotating sensor and the vehicle reliably.

Some embodiments of the present disclosure provide a system. The system includes a rotational bearing configured to enable a platform to rotate, wherein the rotational bearing includes a bearing waveguide having a first end and a second end. The system also includes a vehicle-mounted communication unit. The vehicle-mounted communication unit includes a first set of one or more first communication chips including a first antenna. The vehicle-mounted communication unit also includes at least one interface waveguide configured to couple first electromagnetic signals to and from the first antenna. Additionally, the vehicle-mounted communication unit includes a first waveguide section having a first distal end bordering the first end of the bearing waveguide, and a first proximal end to which the at least one interface waveguide is coupled. The system also includes a platform-mounted communication unit. The platform-mounted communication unit includes a second set of one or more communication chips including a second antenna. The platform-mounted communication unit also includes at least one interface waveguide configured to couple second electromagnetic signals to and from the second antenna. Additionally, the platform-mounted communication unit includes a second waveguide section having a first distal end bordering the second end of the bearing waveguide, and a first proximal end to which the at least one interface waveguide is coupled. Further, the rotational bearing of the system is configured to allow the platform-mounted communication unit to rotate with respect to the vehicle-mounted communication unit, and the rotary joint allows the first and second electromagnetic signals to propagate between the platform-mounted communication unit and the vehicle-mounted communication unit.

Some embodiments of the present disclosure provide a method. A method includes transmitting, by a first antenna of a first set of one or more communication chips, into a first interface waveguide of a first plurality of waveguides of a first electrical coupling, first electromagnetic signals. The method also includes transmitting, by a second antenna of the first set of one or more communication chips, into a second interface waveguide of the first plurality of waveguides of the first electrical coupling, second electromagnetic signals. The method further includes coupling, by the first plurality of waveguides, the first and second electromagnetic signals into a first waveguide section, where the first waveguide section includes a first distal end bordering a bearing waveguide of a rotary joint, a first proximal end to which the first plurality of interface waveguides are coupled, and a first septum. Additionally, the method includes inducing, by the first septum, a respective mode into each of the first and second electromagnetic signals from the first plurality of interface waveguides, where a first mode of the respective modes is orthogonal to the second mode of the respective modes. Moreover, the method includes coupling the first and second electromagnetic signals from the first waveguide section to the bearing waveguide section of the rotary joint, where the bearing waveguide section is part of a rotary joint, and the bearing waveguide section comprises a first end coupled to the first waveguide section and a second end coupled to a second waveguide section. Yet further, the method includes coupling the first and second electromagnetic signals from the bearing waveguide section to a second waveguide section, where the second waveguide section is part of a second electrical coupling and includes a second distal end, a second proximal end to which a second plurality of interface waveguides are coupled, and a second septum. In addition, the method includes dividing, by the second septum, the first and second electromagnetic signals

3

received from the bearing waveguide section to the second plurality of interface waveguides, where dividing the first and second electromagnetic signals to the second plurality of interface waveguides comprises (i) coupling a first subset of the first and second electromagnetic signals into a third interface waveguide of the second plurality of interface waveguides such that the first subset of the first and second electromagnetic signals is coupled having the first mode and (ii) coupling a second subset of the first and second electromagnetic signals into a fourth interface waveguide of the second plurality of interface waveguides such that the second subset of the first and second electromagnetic signals is coupled having the second mode that is orthogonal to the first mode. And, the method also includes coupling, by the second plurality of waveguides, the first and second subsets of the first and second electromagnetic signals to a third antenna of a second set of one or more communication chips and a fourth antenna of the second set of one or more communication chips. Furthermore, the method includes the rotary joint being configured to allow the first electrical coupling to rotate with respect to the second electrical coupling.

Some embodiments of the present disclosure provide a vehicle. The vehicle includes a sensor unit comprising a first set of one or more communication chips including a first antenna and a second antenna. The vehicle also includes a second set of one or more communication chips disposed at a location different from the sensor unit, including a third antenna and a fourth antenna, where the first set of one or more communication chips and the second set of one or more communication chips are configured to engage in two-way communication with each other. The vehicle further includes a rotary joint, having a bearing waveguide. The vehicle also includes a first electrical coupling. The first electrical coupling includes a first plurality of interface waveguides including (i) a first interface waveguide configured to couple first electromagnetic signals to and from the first antenna and (ii) a second interface waveguide configured to couple second electromagnetic signals to and from the second antenna. The first electrical coupling also includes a first waveguide section. The first waveguide section includes a first distal end bordering the bearing waveguide, a first proximal end to which the first plurality of interface waveguides are coupled, and a first septum configured to induce a respective mode into each of the first and second electromagnetic signals from the first plurality of interface waveguides, where a first mode of the respective modes is orthogonal to the second mode of the respective modes. The vehicle further includes a second electrical coupling. The second electrical coupling includes a second plurality of interface waveguides including (i) a third interface waveguide configured to couple third electromagnetic signals to and from the third antenna and (ii) a fourth interface waveguide configured to couple fourth electromagnetic signals to and from the fourth antenna. The second electrical coupling also includes a second waveguide section. The second waveguide section includes a second distal end bordering the bearing waveguide, a second proximal end to which the second plurality of interface waveguides are coupled, and a second septum configured to divide the first and second electromagnetic signals received from the bearing waveguide section, where dividing the first and second electromagnetic signals to the second plurality of interface waveguides comprises (i) coupling a first subset of the first and second electromagnetic signals into a third interface waveguide of the second plurality of interface waveguides such that the first subset of the first and second

4

electromagnetic signals is coupled having the first mode and (ii) coupling a second subset of the first and second electromagnetic signals into a fourth interface waveguide of the second plurality of interface waveguides such that the second subset of the first and second electromagnetic signals is coupled having the second mode that is orthogonal to the first mode. The rotary joint of the vehicle is configured to allow the first electrical coupling to rotate with respect to the second electrical coupling, and wherein the rotary joint allows the first, second, third, and fourth electromagnetic signals to propagate between the first waveguide section and the second waveguide section.

These as well as other aspects, advantages, and alternatives, will become apparent to those of ordinary skill in the art by reading the following detailed description, with reference where appropriate to the accompanying drawings.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a functional block diagram depicting aspects of an example autonomous vehicle.

FIG. 2 depicts exterior views of an example autonomous vehicle.

FIG. 3 illustrates an example waveguide system that forms a contactless electrical coupling.

FIG. 4A illustrates an example microchip having an antenna.

FIG. 4B illustrates an example microchip having two antennas.

FIG. 5 illustrates an example septum of a waveguide.

FIG. 6 illustrates another example waveguide system.

FIG. 7 illustrates an example method.

DETAILED DESCRIPTION

I. Overview

It can be desirable to provide communication of signals to and from the various sensors of the vehicle. For example, one or more sensors may be mounted on the roof of the vehicle. During the operation of the sensor, the sensor may be rotated (e.g., 360°) about a vertical axis, such as by way of being mounted on a rotating platform. Although the sensor and platform are rotating, it may be desirable for the sensor to be in data communication with components on the vehicle, such as a sensor processor. Therefore, it may be desirable to have a system that can reliably communicate signals between the rotating sensor and the non-rotating components.

The rotation of the platform device may present challenges in transmitting communications to, and receiving communications from the respectively rotatable sensor. In particular, it may be undesirable to use cables to transmit communications to, and/or receive communications from the rotatable sensor because, for example, the cables may suffer damage (e.g., due to friction) or become entangled during the rotation of the rotatable sensor.

Disclosed are contactless electrical couplings configured to transmit communications to, and receive communications from a rotatable sensor. The contactless electrical couplings may include a vehicle electrical coupling configured to be mounted on a vehicle and a sensor-side electrical coupling electrically coupled to a rotatable sensor. The contactless electrical couplings may be configured to communicate radio-frequency communications. In some examples, the radio-frequency communications may take the form of electromagnetic energy having a wavelength between 50 and

100 Gigahertz (GHz). In various other examples, the electromagnetic energy may have different frequencies.

The non-rotational side electrical coupling may include (i) at least one communication chip, (ii) at least one interface waveguide, (iii) a first septum, and (iv) a first waveguide section. Similarly, the sensor-side electrical coupling may include (i) at least one communication chip, (ii) at least one interface waveguide, (iii) a second septum, and (iv) a second waveguide section. The two sections may be communicably coupled by way of a waveguide section mounted within a bearing. The bearing may facilitate the rotation of the sensor platform. In order to transmit communications between the two sections, the two waveguide sections and the bearing waveguide section may form a rotary joint.

Herein, a “rotary joint” may refer to a mechanism (or lack thereof) that enables one section of the waveguide to rotate with respect to the other section, and also enables electromagnetic energy to propagate down the length of the waveguide between the two sections, without resulting in any undesirable loss. In essence, the rotary joint electrically couples the two waveguide sections by way of the bearing waveguide section. In some examples, the rotary joint may take the form of two (or more) air gaps (e.g., an air gap between respective ends of the waveguide sections equaling approximately 2 millimeters (mm)).

In practice, one portion of the present waveguide system may be mounted to the vehicle, in communication with a sensor processor, while another portion is mounted to the sensor unit, in communication with a sensor, and a third waveguide is mounted within a bearing section. In some examples, the portion mounted to the vehicle may be integrated within the vehicle itself. In some other examples, the portion mounted to the vehicle may be coupled to the outside of the vehicle, such as a removable sensor unit that can be coupled to the vehicle. For example, the present system may be a single unit that can be connected to a vehicle to provide sensor functionality, thus, the full system may be coupled to a position on the vehicle, such as a roof.

When the sensor unit is mounted the vehicle, the three portions of the waveguides may be brought proximate to each respective end of the bearing waveguide, forming the air gap between each waveguide and the bearing waveguide. During the operation of the waveguide system, vibrations and the rotation of the sensor units may cause the spacing of the air gap and the alignment of the waveguide sections to change. The present system allows for some movement of the various waveguide sections with respect to one another, while maintaining functionality.

As another example, the rotary joint may take the form of a dielectric waveguide or other component configured to couple between two waveguide sections and support rotation of one or both sections around a vertical axis or axes. In such examples, the dielectric waveguide or other component may be configured to align the two sections (e.g., aligned such that the same vertical axis passes through the centers of both sections). However, in these and other examples, there may be scenarios in which the two sections may not be aligned. For instance, the waveguide system may reliably operate with the two centers having a misalignment up to a maximum of approximately 1 mm, or perhaps another maximum in a different implementation.

The waveguide sections may take various forms. In some embodiments, for instance, the waveguide sections may be circular waveguide sections, or another shape of waveguide sections. In other embodiments, the waveguide sections may be square waveguide sections, rectangular waveguide sec-

tions, or another type of polygonal-shaped waveguide sections. Other waveguide section shapes are possible as well.

When the sections are aligned, the rotation of one waveguide section with respect to the other may be rotation around a central axis of the waveguide. However, in some implementations, one section may rotate with respect to the other section without the two sections being aligned. Although the present system will be described as having a vehicle side and a sensor side, in practice, the system may be reciprocal. A reciprocal system will behave similarly when operating forward and backward. Therefore, the terms “vehicle side”, “sensor side”, “transmission”, and “reception” may be used interchangeably in various examples.

During the operation of the waveguide system, an electromagnetic signal may be created by a communication chip. The communication chip may include an integrated antenna. This antenna transmits the electromagnetic signal outside of the chip. This transmitted signal may be coupled into an interface waveguide. The interface waveguide may be designed to efficiently couple signals to and from the communication chip. The interface waveguide may be further configured to couple the electromagnetic signal into a waveguide. The waveguide may include a septum configured to launch a propagation mode in the waveguide. The propagation mode may cause the electromagnetic signal to propagate down the length of a waveguide. The waveguide may have three sections. The middle section of the three sections may be located in a bearing section of a rotary joint.

After the electromagnetic energy crosses the rotary joint, it may encounter a second septum. The second septum may cause the propagation mode to couple the electromagnetic energy into a second interface waveguide. The second interface waveguide may couple the electromagnetic energy out of the second interface waveguide into an antenna located within another communication chip. Therefore, the two communication chips may be in communication with each other by way of the rotary joint and the waveguides. The present system may have high isolation between the input ports of the various interface waveguides. In practice, if a signal is injected into first interface waveguide of the vehicle side (or the sensor side), the other interface waveguide on the same side will see none of (or a very small percentage) of the signal injected into the interface waveguide. Thus, there is a very small or non-existent signal “spillover” from one interface waveguide to the other interface waveguide on the same side of the rotary joint.

An example autonomous vehicle is described below in connection with FIGS. 1-2, while an example rotatable waveguide system is described below in connection with FIGS. 3-7.

II. Example Autonomous Vehicle System

In example embodiments, an example autonomous vehicle system may include one or more processors, one or more forms of memory, one or more input devices/interfaces, one or more output devices/interfaces, and machine-readable instructions that when executed by the one or more processors cause the system to carry out the various functions, tasks, capabilities, etc., described above.

Example systems within the scope of the present disclosure will be described in greater detail below. An example system may be implemented in, or may take the form of, an automobile. However, an example system may also be implemented in or take the form of other vehicles, such as cars, trucks, motorcycles, buses, boats, airplanes, helicopters, lawn mowers, earth movers, boats, snowmobiles, aircraft, recreational vehicles, amusement park vehicles, farm

equipment, construction equipment, trams, golf carts, trains, and trolleys. Other vehicles are possible as well.

FIG. 1 is a functional block diagram illustrating a vehicle 100 according to an example embodiment. The vehicle 100 is configured to operate fully or partially in an autonomous mode, and thus may be referred to as an “autonomous vehicle.” For example, a computer system 112 can control the vehicle 100 while in an autonomous mode via control instructions to a control system 106 for the vehicle 100. The computer system 112 can receive information from one or more sensor systems 104, and base one or more control processes (such as setting a heading so as to avoid a detected obstacle) upon the received information in an automated fashion.

The autonomous vehicle 100 can be fully autonomous or partially autonomous. In a partially autonomous vehicle some functions can optionally be manually controlled (e.g., by a driver) some or all of the time. Further, a partially autonomous vehicle can be configured to switch between a fully-manual operation mode and a partially-autonomous and/or a fully-autonomous operation mode.

The vehicle 100 includes a propulsion system 102, a sensor system 104, a control system 106, one or more peripherals 108, a power supply 110, a computer system 112, and a user interface 116. The vehicle 100 may include more or fewer subsystems and each subsystem can optionally include multiple components. Further, each of the subsystems and components of vehicle 100 can be interconnected and/or in communication. Thus, one or more of the functions of the vehicle 100 described herein can optionally be divided between additional functional or physical components, or combined into fewer functional or physical components. In some further examples, additional functional and/or physical components may be added to the examples illustrated by FIG. 1.

The propulsion system 102 can include components operable to provide powered motion to the vehicle 100. In some embodiments the propulsion system 102 includes an engine/motor 118, an energy source 119, a transmission 120, and wheels/tires 121. The engine/motor 118 converts energy source 119 to mechanical energy. In some embodiments, the propulsion system 102 can optionally include one or both of engines and/or motors. For example, a gas-electric hybrid vehicle can include both a gasoline/diesel engine and an electric motor.

The energy source 119 represents a source of energy, such as electrical and/or chemical energy, that may, in full or in part, power the engine/motor 118. That is, the engine/motor 118 can be configured to convert the energy source 119 to mechanical energy to operate the transmission. In some embodiments, the energy source 119 can include gasoline, diesel, other petroleum-based fuels, propane, other compressed gas-based fuels, ethanol, solar panels, batteries, capacitors, flywheels, regenerative braking systems, and/or other sources of electrical power, etc. The energy source 119 can also provide energy for other systems of the vehicle 100.

The transmission 120 includes appropriate gears and/or mechanical elements suitable to convey the mechanical power from the engine/motor 118 to the wheels/tires 121. In some embodiments, the transmission 120 includes a gearbox, a clutch, a differential, a drive shaft, and/or axle(s), etc.

The wheels/tires 121 are arranged to stably support the vehicle 100 while providing frictional traction with a surface, such as a road, upon which the vehicle 100 moves. Accordingly, the wheels/tires 121 are configured and arranged according to the nature of the vehicle 100. For example, the wheels/tires can be arranged as a unicycle,

bicycle, motorcycle, tricycle, or car/truck four-wheel format. Other wheel/tire geometries are possible, such as those including six or more wheels. Any combination of the wheels/tires 121 of vehicle 100 may be operable to rotate differentially with respect to other wheels/tires 121. The wheels/tires 121 can optionally include at least one wheel that is rigidly attached to the transmission 120 and at least one tire coupled to a rim of a corresponding wheel that makes contact with a driving surface. The wheels/tires 121 may include any combination of metal and rubber, and/or other materials or combination of materials.

The sensor system 104 generally includes one or more sensors configured to detect information about the environment surrounding the vehicle 100. For example, the sensor system 104 can include a Global Positioning System (GPS) 122, an inertial measurement unit (IMU) 124, a RADAR unit 126, a laser rangefinder/LIDAR unit 128, a camera 130, and/or a microphone 131. The sensor system 104 could also include sensors configured to monitor internal systems of the vehicle 100 (e.g., O₂ monitor, fuel gauge, engine oil temperature, wheel speed sensors, etc.). One or more of the sensors included in sensor system 104 could be configured to be actuated separately and/or collectively in order to modify a position and/or an orientation of the one or more sensors.

The GPS 122 is a sensor configured to estimate a geographic location of the vehicle 100. To this end, GPS 122 can include a transceiver operable to provide information regarding the position of the vehicle 100 with respect to the Earth.

The IMU 124 can include any combination of sensors (e.g., accelerometers and gyroscopes) configured to sense position and orientation changes of the vehicle 100 based on inertial acceleration.

The RADAR unit 126 can represent a system that utilizes radio signals to sense objects within the local environment of the vehicle 100. In some embodiments, in addition to sensing the objects, the RADAR unit 126 and/or the computer system 112 can additionally be configured to sense the speed and/or heading of the objects.

Similarly, the laser rangefinder or LIDAR unit 128 can be any sensor configured to sense objects in the environment in which the vehicle 100 is located using lasers. The laser rangefinder/LIDAR unit 128 can include one or more laser sources, a laser scanner, and one or more detectors, among other system components. The laser rangefinder/LIDAR unit 128 can be configured to operate in a coherent (e.g., using heterodyne detection) or an incoherent detection mode.

The camera 130 can include one or more devices configured to capture a plurality of images of the environment surrounding the vehicle 100. The camera 130 can be a still camera or a video camera. In some embodiments, the camera 130 can be mechanically movable such as by rotating and/or tilting a platform to which the camera is mounted. As such, a control process of vehicle 100 may be implemented to control the movement of camera 130.

The sensor system 104 can also include a microphone 131. The microphone 131 can be configured to capture sound from the environment surrounding vehicle 100. In some cases, multiple microphones can be arranged as a microphone array, or possibly as multiple microphone arrays.

The control system 106 is configured to control operation(s) regulating acceleration of the vehicle 100 and its components. To effect acceleration, the control system 106 includes a steering unit 132, throttle 134, brake unit 136, a

sensor fusion algorithm 138, a computer vision system 140, a navigation/pathing system 142, and/or an obstacle avoidance system 144, etc.

The steering unit 132 is operable to adjust the heading of vehicle 100. For example, the steering unit can adjust the axis (or axes) of one or more of the wheels/tires 121 so as to effect turning of the vehicle. The throttle 134 is configured to control, for instance, the operating speed of the engine/motor 118 and, in turn, adjust forward acceleration of the vehicle 100 via the transmission 120 and wheels/tires 121. The brake unit 136 decelerates the vehicle 100. The brake unit 136 can use friction to slow the wheels/tires 121. In some embodiments, the brake unit 136 inductively decelerates the wheels/tires 121 by a regenerative braking process to convert kinetic energy of the wheels/tires 121 to electric current.

The sensor fusion algorithm 138 is an algorithm (or a computer program product storing an algorithm) configured to accept data from the sensor system 104 as an input. The data may include, for example, data representing information sensed at the sensors of the sensor system 104. The sensor fusion algorithm 138 can include, for example, a Kalman filter, Bayesian network, etc. The sensor fusion algorithm 138 provides assessments regarding the environment surrounding the vehicle based on the data from sensor system 104. In some embodiments, the assessments can include evaluations of individual objects and/or features in the environment surrounding vehicle 100, evaluations of particular situations, and/or evaluations of possible interference between the vehicle 100 and features in the environment (e.g., such as predicting collisions and/or impacts) based on the particular situations.

The computer vision system 140 can process and analyze images captured by camera 130 to identify objects and/or features in the environment surrounding vehicle 100. The detected features/objects can include traffic signals, roadway boundaries, other vehicles, pedestrians, and/or obstacles, etc. The computer vision system 140 can optionally employ an object recognition algorithm, a Structure From Motion (SFM) algorithm, video tracking, and/or available computer vision techniques to effect categorization and/or identification of detected features/objects. In some embodiments, the computer vision system 140 can be additionally configured to map the environment, track perceived objects, estimate the speed of objects, etc.

The navigation and pathing system 142 is configured to determine a driving path for the vehicle 100. For example, the navigation and pathing system 142 can determine a series of speeds and directional headings to effect movement of the vehicle along a path that substantially avoids perceived obstacles while generally advancing the vehicle along a roadway-based path leading to an ultimate destination, which can be set according to user inputs via the user interface 116, for example. The navigation and pathing system 142 can additionally be configured to update the driving path dynamically while the vehicle 100 is in operation on the basis of perceived obstacles, traffic patterns, weather/road conditions, etc. In some embodiments, the navigation and pathing system 142 can be configured to incorporate data from the sensor fusion algorithm 138, the GPS 122, and one or more predetermined maps so as to determine the driving path for vehicle 100.

The obstacle avoidance system 144 can represent a control system configured to identify, evaluate, and avoid or otherwise negotiate potential obstacles in the environment surrounding the vehicle 100. For example, the obstacle avoidance system 144 can effect changes in the navigation

of the vehicle by operating one or more subsystems in the control system 106 to undertake swerving maneuvers, turning maneuvers, braking maneuvers, etc. In some embodiments, the obstacle avoidance system 144 is configured to automatically determine feasible (“available”) obstacle avoidance maneuvers on the basis of surrounding traffic patterns, road conditions, etc. For example, the obstacle avoidance system 144 can be configured such that a swerving maneuver is not undertaken when other sensor systems detect vehicles, construction barriers, other obstacles, etc. in the region adjacent the vehicle that would be swerved into. In some embodiments, the obstacle avoidance system 144 can automatically select the maneuver that is both available and maximizes safety of occupants of the vehicle. For example, the obstacle avoidance system 144 can select an avoidance maneuver predicted to cause the least amount of acceleration in a passenger cabin of the vehicle 100.

The vehicle 100 also includes peripherals 108 configured to allow interaction between the vehicle 100 and external sensors, other vehicles, other computer systems, and/or a user, such as an occupant of the vehicle 100. For example, the peripherals 108 for receiving information from occupants, external systems, etc. can include a wireless communication system 146, a touchscreen 148, a microphone 150, and/or a speaker 152.

In some embodiments, the peripherals 108 function to receive inputs for a user of the vehicle 100 to interact with the user interface 116. To this end, the touchscreen 148 can both provide information to a user of vehicle 100, and convey information from the user indicated via the touchscreen 148 to the user interface 116. The touchscreen 148 can be configured to sense both touch positions and touch gestures from a user’s finger (or stylus, etc.) via capacitive sensing, resistance sensing, optical sensing, a surface acoustic wave process, etc. The touchscreen 148 can be capable of sensing finger movement in a direction parallel or planar to the touchscreen surface, in a direction normal to the touchscreen surface, or both, and may also be capable of sensing a level of pressure applied to the touchscreen surface. An occupant of the vehicle 100 can also utilize a voice command interface. For example, the microphone 150 can be configured to receive audio (e.g., a voice command or other audio input) from a user of the vehicle 100. Similarly, the speakers 152 can be configured to output audio to the user of the vehicle 100.

In some embodiments, the peripherals 108 function to allow communication between the vehicle 100 and external systems, such as devices, sensors, other vehicles, etc. within its surrounding environment and/or controllers, servers, etc., physically located far from the vehicle that provide useful information regarding the vehicle’s surroundings, such as traffic information, weather information, etc. For example, the wireless communication system 146 can wirelessly communicate with one or more devices directly or via a communication network. The wireless communication system 146 can optionally use 3G cellular communication, such as—Code-Division Multiple Access (CDMA), Evolution-Data Optimized (EV-DO), Global System for Mobile communications (GSM)/General Packet Radio Surface (GPRS), and/or 4G cellular communication, such as Worldwide Interoperability for Microwave Access (WiMAX) or Long-Term Evolution (LTE). Additionally or alternatively, wireless communication system 146 can communicate with a wireless local area network (WLAN), for example, using WiFi. In some embodiments, wireless communication system 146 could communicate directly with a device, for example, using an infrared link, Bluetooth®, and/or Zig-

11

Bee®. The wireless communication system **146** can include one or more dedicated short-range communication (DSRC) devices that can include public and/or private data communications between vehicles and/or roadside stations. Other wireless protocols for sending and receiving information embedded in signals, such as various vehicular communication systems, can also be employed by the wireless communication system **146** within the context of the present disclosure.

As noted above, the power supply **110** can provide power to components of vehicle **100**, such as electronics in the peripherals **108**, computer system **112**, sensor system **104**, etc. The power supply **110** can include a rechargeable lithium-ion or lead-acid battery for storing and discharging electrical energy to the various powered components, for example. In some embodiments, one or more banks of batteries can be configured to provide electrical power. In some embodiments, the power supply **110** and energy source **119** can be implemented together, as in some all-electric cars.

Many or all of the functions of vehicle **100** can be controlled via computer system **112** that receives inputs from the sensor system **104**, peripherals **108**, etc., and communicates appropriate control signals to the propulsion system **102**, control system **106**, peripherals **108**, etc. to effect automatic operation of the vehicle **100** based on its surroundings. Computer system **112** includes at least one processor **113** (which can include at least one microprocessor) that executes instructions **115** stored in a non-transitory computer readable medium, such as the data storage **114**. The computer system **112** may also represent a plurality of computing devices that serve to control individual components or subsystems of the vehicle **100** in a distributed fashion.

In some embodiments, data storage **114** contains instructions **115** (e.g., program logic) executable by the processor **113** to execute various functions of vehicle **100**, including those described above in connection with FIG. 1. Data storage **114** may contain additional instructions as well, including instructions to transmit data to, receive data from, interact with, and/or control one or more of the propulsion system **102**, the sensor system **104**, the control system **106**, and the peripherals **108**.

In addition to the instructions **115**, the data storage **114** may store data such as roadway maps, path information, among other information. Such information may be used by vehicle **100** and computer system **112** during operation of the vehicle **100** in the autonomous, semi-autonomous, and/or manual modes to select available roadways to an ultimate destination, interpret information from the sensor system **104**, etc.

The vehicle **100**, and associated computer system **112**, provides information to and/or receives input from, a user of vehicle **100**, such as an occupant in a passenger cabin of the vehicle **100**. The user interface **116** can accordingly include one or more input/output devices within the set of peripherals **108**, such as the wireless communication system **146**, the touchscreen **148**, the microphone **150**, and/or the speaker **152** to allow communication between the computer system **112** and a vehicle occupant.

The computer system **112** controls the operation of the vehicle **100** based on inputs received from various subsystems indicating vehicle and/or environmental conditions (e.g., propulsion system **102**, sensor system **104**, and/or control system **106**), as well as inputs from the user interface **116**, indicating user preferences. For example, the computer system **112** can utilize input from the control system **106** to

12

control the steering unit **132** to avoid an obstacle detected by the sensor system **104** and the obstacle avoidance system **144**. The computer system **112** can be configured to control many aspects of the vehicle **100** and its subsystems. Generally, however, provisions are made for manually overriding automated controller-driven operation, such as in the event of an emergency, or merely in response to a user-activated override, etc.

The components of vehicle **100** described herein can be configured to work in an interconnected fashion with other components within or outside their respective systems. For example, the camera **130** can capture a plurality of images that represent information about an environment of the vehicle **100** while operating in an autonomous mode. The environment may include other vehicles, traffic lights, traffic signs, road markers, pedestrians, etc. The computer vision system **140** can categorize and/or recognize various aspects in the environment in concert with the sensor fusion algorithm **138**, the computer system **112**, etc. based on object recognition models pre-stored in data storage **114**, and/or by other techniques.

Although the vehicle **100** is described and shown in FIG. 1 as having various components of vehicle **100**, e.g., wireless communication system **146**, computer system **112**, data storage **114**, and user interface **116**, integrated into the vehicle **100**, one or more of these components can optionally be mounted or associated separately from the vehicle **100**. For example, data storage **114** can exist, in part or in full, separate from the vehicle **100**, such as in a cloud-based server, for example. Thus, one or more of the functional elements of the vehicle **100** can be implemented in the form of device elements located separately or together. The functional device elements that make up vehicle **100** can generally be communicatively coupled together in a wired and/or wireless fashion.

FIG. 2 shows an example vehicle **200** that can include some or all of the functions described in connection with vehicle **100** in reference to FIG. 1. In particular, FIG. 2 shows various different views of vehicle **200**, labeled in FIG. 2 as “Right Side View,” “Front View,” “Back View,” and “Top View.” Although vehicle **200** is illustrated in FIG. 2 as a four-wheel van-type car for illustrative purposes, the present disclosure is not so limited. For instance, the vehicle **200** can represent a truck, a van, a semi-trailer truck, a motorcycle, a golf cart, an off-road vehicle, or a farm vehicle, etc.

The example vehicle **200** includes a sensor unit **202**, a wireless communication system **204**, RADAR units **206**, laser rangefinder units **208**, and a camera **210**. Furthermore, the example vehicle **200** can include any of the components described in connection with vehicle **100** of FIG. 1. The RADAR unit **206** and/or laser rangefinder unit **208** can actively scan the surrounding environment for the presence of potential obstacles and can be similar to the RADAR unit **126** and/or laser rangefinder/LIDAR unit **128** in the vehicle **100** of FIG. 1.

The sensor unit **202** is mounted atop the vehicle **200** and includes one or more sensors configured to detect information about an environment surrounding the vehicle **200**, and output indications of the information. For example, sensor unit **202** can include any combination of cameras, RADARs, LIDARs, range finders, and acoustic sensors. The sensor unit **202** can include one or more movable mounts that could be operable to adjust the orientation of one or more sensors in the sensor unit **202**. In one embodiment, the movable mount could include a rotating platform that could scan sensors so as to obtain information from each direction

around the vehicle **200**. In another embodiment, the movable mount of the sensor unit **202** could be moveable in a scanning fashion within a particular range of angles and/or azimuths. The sensor unit **202** could be mounted atop the roof of a car, for instance, however other mounting locations are possible. Additionally, the sensors of sensor unit **202** could be distributed in different locations and need not be collocated in a single location. Some possible sensor types and mounting locations include RADAR unit **206** and laser rangefinder unit **208**. Furthermore, each sensor of sensor unit **202** can be configured to be moved or scanned independently of other sensors of sensor unit **202**.

In an example configuration, one or more RADAR scanners (e.g., the RADAR unit **206**) can be located near the front of the vehicle **200**, to actively scan the region in front of the car **200** for the presence of radio-reflective objects. A RADAR scanner can be situated, for example, in a location suitable to illuminate a region including a forward-moving path of the vehicle **200** without occlusion by other features of the vehicle **200**. For example, a RADAR scanner can be situated to be embedded and/or mounted in or near the front bumper, front headlights, cowl, and/or hood, etc. Furthermore, one or more additional RADAR scanning devices can be located to actively scan the side and/or rear of the vehicle **200** for the presence of radio-reflective objects, such as by including such devices in or near the rear bumper, side panels, rocker panels, and/or undercarriage, etc.

The wireless communication system **204** could be located on a roof of the vehicle **200** as depicted in FIG. 2. Alternatively, the wireless communication system **204** could be located, fully or in part, elsewhere. The wireless communication system **204** may include wireless transmitters and receivers that could be configured to communicate with devices external or internal to the vehicle **200**. Specifically, the wireless communication system **204** could include transceivers configured to communicate with other vehicles and/or computing devices, for instance, in a vehicular communication system or a roadway station. Examples of such vehicular communication systems include dedicated short range communications (DSRC), radio frequency identification (RFID), and other proposed communication standards directed towards intelligent transport systems.

The camera **210** can be a photo-sensitive instrument, such as a still camera, a video camera, etc., that is configured to capture a plurality of images of the environment of the vehicle **200**. To this end, the camera **210** can be configured to detect visible light, and can additionally or alternatively be configured to detect light from other portions of the spectrum, such as infrared or ultraviolet light. The camera **210** can be a two-dimensional detector, and can optionally have a three-dimensional spatial range of sensitivity. In some embodiments, the camera **210** can include, for example, a range detector configured to generate a two-dimensional image indicating distance from the camera **210** to a number of points in the environment. To this end, the camera **210** may use one or more range detecting techniques.

For example, the camera **210** can provide range information by using a structured light technique in which the vehicle **200** illuminates an object in the environment with a predetermined light pattern, such as a grid or checkerboard pattern and uses the camera **210** to detect a reflection of the predetermined light pattern from environmental surroundings. Based on distortions in the reflected light pattern, the vehicle **200** can determine the distance to the points on the object. The predetermined light pattern may comprise infrared light, or radiation at other suitable wavelengths for such measurements.

The camera **210** can be mounted inside a front windshield of the vehicle **200**. Specifically, the camera **210** can be situated to capture images from a forward-looking view with respect to the orientation of the vehicle **200**. Other mounting locations and viewing angles of camera **210** can also be used, either inside or outside the vehicle **200**. Further, the camera **210** can have associated optics operable to provide an adjustable field of view. Further, the camera **210** can be mounted to vehicle **200** with a movable mount to vary a pointing angle of the camera **210**, such as via a pan/tilt mechanism.

III. Example Communication System

In some cases, the sensor unit **202** described above in connection with FIG. 2 may include a variety of sensors, such as LIDAR, RADAR, other optical sensors, and/or other sensors. During the operation of the sensors, it may be desirable to communicate a large amount of data between the sensor unit and various systems of the vehicle, such as a processor for processing the data from the sensor.

The sensor unit may include a rotatable set of sensors that is configured to rotate (e.g., 360°) about a vertical axis. Further, the rotatable sensor device may include contactless electrical couplings configured to transmit communications to, and receive communications from the rotatable sensor unit. For example, the communications may be data sent to or received from the sensor unit.

FIG. 3 illustrates an example waveguide system **300** that forms the contactless electrical coupling. The rotary joint **304** may enable the rotation of the sensor platform when it is mounted to the vehicle. As previously discussed, during the operation of a sensor system, it may be desirable for the sensor system to rotate. However, the rotation may cause difficulty in the communication of signals between the sensor (which is rotating) and a processing system (which is mounted on a fixed platform, such as a vehicle). Thus, the waveguide system **300** may be used to couple signals between the sensor unit and the processing system, or chips communicably coupled to the sensor unit and the processing system. In one example, the rotary joint **304** may include a bearing **318** that enables the rotation of the sensor platform. The bearing **318** may be mounted to the vehicle portion on one side and the sensor platform on the other side. The bearing **318** may enable the sensor platform to be physically coupled to the vehicle in a manner that enables the rotation of the sensor platform. Further, the bearing **318** may include a hollow center portion, into which the present bearing mount section **314** may be inserted. Bearing **318** is one example of a possible bearing for the present system, other shapes, sizes, and configurations of bearings are possible as well.

In one example, the sensor unit may be a LIDAR sensor. The processing system may include a LIDAR processor (not shown). Due to power requirements of the LIDAR processor, it may be desirable to have the LIDAR processor mounted on the vehicle (i.e., the non-rotating portion). The LIDAR processor may be communicably coupled (e.g., connected with wires, fiber optics, dielectrics, etc.) to the LIDAR chip of the rotary joint. The LIDAR chip may communicate a signal across the rotary joint to another LIDAR chip of the sensor unit. This LIDAR chip in the sensor unit may control the operation of the LIDAR sensor. Further, the LIDAR sensor may use this LIDAR chip to communicate LIDAR data back to the vehicle-mounted LIDAR chip for processing by the LIDAR processor. Thus, the present rotary joint system enables the processing system for a given sensor to not be located on the spinning platform

15

with the sensor itself. This may make powering the processor easier, fabrication easier, and also increase mechanical robustness of the system.

Additionally, the bearing **318** may be mechanically coupled to a motor or actuator configured to provide a rotation of the sensor platform. In some examples, a bearing waveguide may rotate along with the sensor platform about an axis of rotation. In other examples, the bearing waveguide may be fixed with respect to the vehicle. In both examples, the rotary joint provides similar function.

The bearing mount section **314** may fit within a center portion of the bearing **318** and provide a bearing waveguide **302C** that can communicate radio frequency signals through the rotary joint **304**. The bearing mount section **314** may be machined to have dimensions that enable it to mount within a center portion of the bearing **318**.

Example waveguide system **300** includes circular waveguide sections as representative waveguide sections, although, as noted above, other types and shapes of waveguide sections are possible in other waveguide systems. In particular, the waveguide system **300** includes a first circular waveguide section **302A**, a second circular waveguide section **302B**, and a bearing waveguide section **302C**. The first circular waveguide section **302A** and the second circular waveguide section **302B** may be electrically coupled by way of the bearing waveguide section **302C**. At the rotary joint **304**, the first circular waveguide section **302A**, the second circular waveguide section **302B**, and the bearing waveguide section **302C** may be approximately aligned based on the center axis of the circular portion of the waveguide.

The waveguides that form system **300** may be constructed of a metallic material, a non-metallic material that has been plated with a metallic surface, a dielectric material, a combination of these materials, or other materials that may have electromagnetic properties to contain and allow the propagation of electromagnetic signals.

In various embodiments, the rotary joint **304** may take various forms. As shown in the figures, the rotary joint **304** may include the bearing waveguide section **302C** and a bearing mount section **314**. The bearing mount section **314** may be approximately a cylindrical shape and be configured to mount to an internal portion of a rotational bearing. Additionally, the bearing mount section **314** may be symmetrical about a plane defined by plane **316**. Further, the bearing mount section **314** may include a flange portion adapted to interface with the first circular waveguide section **302A** and the second circular waveguide section **302B**. The flange portion may help align the first circular waveguide section **302A** and the second circular waveguide section **302B** with the bearing waveguide section **302C**. In some examples, the flange portion may be omitted. Additionally, the first circular waveguide section **302A**, the second circular waveguide section **302B**, and the bearing waveguide section **302C** may each have a similar diameter (or the same diameter, inclusive of machining tolerances).

In some examples, the first circular waveguide section **302A** and the second circular waveguide section **302B** may be separated from the bearing waveguide section **302C** by an air gap on the order of 1-3 millimeter (mm) when the electromagnetic energy is between 50 and 100 Ghz. The air gap does not have to be between 1 mm and 3 mm. In some examples, the air gap may be bigger or smaller. In some other examples, there may not be an air gap, but rather the first circular waveguide section **302A** and the second circular waveguide section **302B** may be in contact with a flange portion of the mount section **314**.

16

In practice, one portion of the present waveguide system may be mounted to the vehicle while the other portion is mounted to the sensor unit. When the sensor unit is mounted to the vehicle, the two portions of the waveguide may be brought proximate to the bearing waveguide section **302C**, forming the air gaps. During the operation of the waveguide system, vibrations and the rotation of the sensor units may cause the spacing of the air gap and the alignment of the various waveguide sections to change.

As previously discussed with respect to other examples, the rotary joint **304** may include a physical connection between the first circular waveguide section **302A**, the second circular waveguide section **302B**, and the bearing waveguide section **302C**, possibly by way of the flanged portion of the flange portion of the mount section **314**. The physical connection may be an abutment of the ends of the first circular waveguide section **302A**, the second circular waveguide section **302B**, and the bearing waveguide section **302C**. In some additional examples, the rotary joint **304** may include other components as well. For example, the rotary joint **304** may include some additional components, such as a bearing sleeve, slip ring, or similar structure, that help align the first circular waveguide section **302A** and the second circular waveguide section **302B** with the bearing waveguide section **302C**, while allowing for rotation.

The first circular waveguide section **302A** may be coupled to a plurality of interface waveguides, shown as interface waveguide **306A** and **306C**. The second circular waveguide section **302B** may also be coupled to a plurality of interface waveguides, shown as interface waveguide **306B** and **306D**.

Each interface waveguide may be coupled to a communication chip having a chip antenna. For example, interface waveguide **306A** is coupled to communication chip **308A** by way of chip antenna **310A**, interface waveguide **306B** is coupled to communication chip **308B** by way of chip antenna **310B**, interface waveguide **306C** is coupled to communication chip **308C** by way of chip antenna **310C**, and interface waveguide **306D** is coupled to communication chip **308D** by way of chip antenna **310D**. In some examples, a single communication chip may have multiple antenna and therefore a single chip may be coupled to multiple interface waveguide (as shown with respect to FIG. 6).

The first circular waveguide section **302A** may include a septum **312A** and the second circular waveguide section **302B** may include a septum **312B**. Each septum may be aligned in a vertical manner on a plane defined by a center of where the interface waveguides couple to the circular waveguide. Essentially, the septums may form a wall in the circular waveguide between the openings of the two interface waveguides.

As so arranged, example waveguide system **300** may in some implementations operate such that communication chips **308A** and **308C** may communicate with communication chips **308B** and **308D** by way of the interface waveguides and the three circular waveguide sections. For example, communication chip **308A** may transmit a signal via antenna **310A**, and communication chip **308C** may transmit a signal via antenna **310C**. The signal from chip **308A** may be coupled into and propagate through interface waveguide **306A**, and the signal from chip **308C** may be coupled into and propagate through interface waveguide **308C**. Each of these interface waveguides may in turn efficiently couple the two signals into circular waveguide section **302A**.

In line with the discussion above, septum **312A** may induce a propagation mode to each of the two signals to have orthogonal modes, after which the two signals may propa-

17

gate down the length of the circular waveguide, through the bearing waveguide section **302C** of the rotary joint, to septum **312B**. The two signals having orthogonal modes at this point may enable septum **312B** to divide the signals based on the respective modes, and, in turn, couple one of the divided signals to interface waveguide **306B**, and couple the other of the divided signals to interface waveguide **306D**. The signals may then propagate through the respective interface waveguides to be coupled into the communication chips **308B** and **308D**, which receives the signals via antennas **310B** and **310D**, respectively.

FIG. **4A** illustrates an example microchip **402** having an antenna **404**. The antenna **404** may be used by the microchip **402** to communicate signals out of and into the microchip **402**. Often, and especially at radio frequencies, the interface to and from a microchip may be inefficient and or difficult to design. Therefore, to improve chip communications, microchips may include antennas that can communicate signals to components external to the microchip.

In conventional systems, an external component may have an antenna that receives the signal output by the antenna of the microchip (or the external antenna can transmit a signal to be received by the antenna of the microchip). The present system uses a waveguide to directly harvest the electromagnetic energy transmitted by the antenna of the microchip. By using a waveguide to harvest the energy, the system may be able to communicate signals from the microchip to other various components in an efficient manner.

FIG. **4B** illustrates an example microchip **452** having two antennas **454A** and **454B**. The example microchip **452** also includes a grounding portion **456** located between the two antennas **454A** and **454B**. Microchip **452** may include two (or more) antennas, each of which functions in a similar manner to the antennas of microchip **402**. Each antenna of microchip **452** may be coupled to a respective interface waveguide. In addition, microchip **452** may have a grounding portion **456**. The grounding portion **456** may be coupled to the waveguide structure disclosed herein. By grounding the grounding portion **456** to the waveguide structure, the two antennas may be sufficiently isolated from each other. When the two antennas are isolated from each other, each antenna may not receive (or receive a small portion of) signals communicated to or from the other respective antenna.

FIG. **5** illustrates an example septum **504** of a waveguide **502**. As shown, the septum may have a stepped pattern. The septum **504** may be constructed of a metallic material, a non-metallic material that has been plated with a metallic surface, a dielectric material, a combination of these materials, or other materials that may have electromagnetic properties to alter electromagnetic signals. The stepped pattern may cause a signal that begins propagation on one side of the septum to have an orthogonal mode to a signal that begins propagation on the other side of the septum. Similarly, the stepped pattern may be able to split electromagnetic energy based on the modes contained in the energy. The stepped pattern may cause a portion signal that has a first mode to continue propagation on one side of the septum and may cause a portion signal that has a second mode to continue propagation on the other side of the septum.

Through the use of the septum separating the propagation modes, two chips who are in communication with each other by way of the present waveguide structure may remain in communication irrespective of the rotation of the waveguides. Therefore, signals sent by an antenna of one chip

18

may be able to be received by the corresponding chip throughout the entire rotation. Although the present septum is shown having the stepped pattern, other shapes may be used as well. In some examples, or where orthogonality is not desired, the septum may be omitted.

FIG. **6** illustrates another example waveguide system **600** that forms the contactless electrical coupling. The waveguide system **600** includes a first circular waveguide section **602A**, a second circular waveguide section **602B**, and a bearing waveguide section **602C**. The first circular waveguide section **602A** and the second circular waveguide section **602B** may be electrically coupled by way of the bearing waveguide section **602C** of the rotary joint **604**.

The first circular waveguide section **602A** may be coupled to a plurality of interface waveguides, shown as interface waveguides **606A** and **606C**. The second circular waveguide section **602B** may also be coupled to a plurality of interface waveguides, shown as interface waveguide **606B** and **606D**.

Each interface waveguide may be coupled to a communication chip having a chip antenna. For example, interface waveguide **606A** is coupled to communication chip **608A** by way of chip antenna **610A**, interface waveguide **606C** is coupled to communication chip **608C** by way of chip antenna **610C**, interface waveguide **606B** is coupled to communication chip **608B** by way of chip antenna **610B**, and interface waveguide **606D** is coupled to communication chip **608D** by way of chip antenna **610D**. The first circular waveguide section **602A** may include a septum **612A** and the second circular waveguide section **602B** may include a septum **612B**.

As so arranged, example waveguide system **600** may operate similarly to example waveguide system **300** described above in FIG. **3** (e.g., signals may propagate through the system and be caused to have orthogonal modes), except with a single communication chip **608A** (such as microchip **452** of FIG. **4B**) transmitting the initial signals via antennas **610A** and **610C** and a single communication chip **608B** receiving the split signals via antennas **610B** and **610D**. For brevity, the bearing **318** (of FIG. **3**) is omitted from FIG. **6**.

Additionally, more chips and antennas may be included as well. Each antenna may be coupled to its own respective interface waveguide. In some examples, there may be four antennas, and four interface waveguides on each side of the waveguide system. Other possible examples are possible as well.

Many variations on the above-described implementations are possible as well, each advantageously and reliably providing communications between the vehicle and at least one sensor. In one implementation, for instance, either the vehicle side or the sensor side may not include a set of receiving chips. For example, the vehicle may include a set of chips configured to transmit signals through a waveguide system similar to that described herein for direct receipt by a radar unit (e.g., RADAR unit **126** of FIG. **1**). As a result, the radar unit may receive signals with less signal loss and/or other changes than if a receiving chip had received the signals and coupled them to the radar unit. The present system may be used with radar units, LIDAR units, camera units, or other sensor units of the vehicle as well.

FIG. **7** illustrates an example method. At block **702**, the method transmitting into a first interface waveguide of a first plurality of waveguides of a first electrical coupling, first electromagnetic signals. The transmitting may be performed by a first antenna of a first set of one or more communication chips, such as a microchip. The first interface waveguide may be designed in a way to attempt to maximize the

19

amount of energy that is transmitted by antenna that couples into the waveguide. The signal may couple into a first end of the first interface waveguide.

At block **704**, the method includes transmitting into a second interface waveguide of the first plurality of waveguides of the first electrical coupling, second electromagnetic signals. The transmitting may be performed by a second antenna of a first set of one or more communication chips, such as a microchip. The first interface waveguide may be designed in a way to attempt to maximize the amount of energy that is transmitted by antenna that couples into the waveguide. The signal may couple into a first end of the second interface waveguide. In some examples, the first set of one or more communication chips includes a single communication chip. In some other examples, the first set of one or more communication chips includes two communication chips.

At block **706**, the method includes coupling the first and second electromagnetic signals into a first waveguide section. The coupling may be performed by the first plurality of waveguides. The first waveguide section includes a first distal end bordering a bearing waveguide of a rotary joint, a first proximal end to which the first plurality of interface waveguides are coupled, and a first septum. The coupling from the interface waveguides may cause the signal received from the microchip to start propagating in the waveguide.

At block **708**, the method includes inducing a respective mode into each of the first and second electromagnetic signals from the first plurality of interface waveguides. The inducing modes may be performed by a the first septum. Additionally, a first mode of the respective modes is orthogonal to the second mode of the respective modes. For example, a signal from an interface waveguide may have a first mode induced by the septum and a signal from another interface waveguide may have a second mode induced by the septum. Because the modes may be orthogonal to each other, the original signals may be retrieved (at a later block) based on the septum dividing the signal based on modes.

At block **710**, the method includes coupling the first and second electromagnetic signals from the first waveguide section to the bearing waveguide section of the rotary joint. The bearing waveguide section may form part of a rotary joint, and the bearing waveguide section includes a first end coupled to the first waveguide section and a second end coupled to a second waveguide section. The rotary joint, including the bearing waveguide, is configured to allow the first electrical coupling to rotate with respect to the second electrical coupling.

At block **712**, the method includes coupling the first and second electromagnetic signals from the bearing waveguide section to a second waveguide section. The second waveguide section may form part of a second electrical coupling and includes a second distal end, a second proximal end to which a second plurality of interface waveguides are coupled, and a second septum.

At block **714**, the method includes dividing the first and second electromagnetic signals received from the bearing waveguide section to the second plurality of interface waveguides. The dividing of block **714** may be performed by a second septum. Dividing the first and second electromagnetic signals to the second plurality of interface waveguides may include (i) coupling a first subset of the first and second electromagnetic signals into a third interface waveguide of the second plurality of interface waveguides such that the first subset of the first and second electromagnetic signals is coupled having the first mode and (ii) coupling a second subset of the first and second electromagnetic signals into a

20

fourth interface waveguide of the second plurality of interface waveguides such that the second subset of the first and second electromagnetic signals is coupled having the second mode that is orthogonal to the first mode.

At block **716**, the method includes coupling, by the second plurality of waveguides, the first and second subsets of the first and second electromagnetic signals to a third antenna of a second set of one or more communication chips and a fourth antenna of the second set of one or more communication chips. In some examples, the second set of one or more communication chips includes a single communication chip. In some other examples, the second set of one or more communication chips includes two communication chips.

Therefore, method **700** enables two microchips to be in radio frequency with each other even while one of the two microchips is mounted on a rotating platform. In some examples, signals may not originate or end at microchips. Other structures may be used to launch or receive signals from the interface waveguides. For example, a radar signal generator and receiver may be coupled to one interface waveguide. On the other end may be a radar antenna. By way of the present system, the radar antenna may be on a rotating platform while maintaining communication with the radar signal generator and receiver.

In some other examples, there may be multiple interface waveguides on each end of the waveguides. In these examples, due to creating orthogonal signals, multiple signals may be communicated through the rotary joint and recovered separately after the rotary joint. In some other examples, the radiating may be performed in both directions simultaneously, with one signal propagating in one direction and another signal propagating in the other direction. When transmitting signals in both directions simultaneously, the system may operate in a full-duplex mode with two different channels.

In some further examples of the method, the first set of one or more communication chips is coupled to a sensor unit attached to a vehicle and the second set of one or more communication chips is coupled to the vehicle at a location different from the sensor unit, and as previously discussed, the communication system enables two-way communication between the vehicle and the sensor unit. In some examples, the first set of one or more communication chips are part of a light detection and ranging (LIDAR) sensor included in the sensor unit.

In some further additional examples of the method, the method may also include providing a rotation of the bearing waveguide with respect to the first waveguide section and/or providing a rotation of the bearing waveguide with respect to the second waveguide section. Additionally, the first and second electromagnetic signals of the method may have a frequency between 50 and 100 Gigahertz.

While various example aspects and example embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various example aspects and example embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

What is claimed is:

1. A light detection and ranging (LIDAR) system comprising:

- a non-rotational unit having a first waveguide section, wherein the non-rotational unit is coupled in a fixed position to a vehicle;
- a rotational unit having a second waveguide section; and
- a rotary joint having a bearing waveguide located in a center portion of a rotational bearing of the rotary joint,

21

wherein the rotational bearing is configured to allow the rotational unit to rotate with respect to the non-rotational unit,

wherein the bearing waveguide is aligned with the first waveguide section and the second waveguide section such that electromagnetic signals are able to propagate between the rotational unit and the non-rotational unit, and

wherein the rotational unit includes a LIDAR sensor, and wherein the electromagnetic signals comprise LIDAR data from the LIDAR sensor.

2. The LIDAR system of claim 1, wherein the electromagnetic signals further comprise control signals for controlling the LIDAR sensor.

3. The LIDAR system of claim 1, wherein the LIDAR sensor comprises one or more laser sources and one or more detectors.

4. The LIDAR system of claim 3, wherein the LIDAR sensor further comprises a laser scanner.

5. The LIDAR system of claim 1, wherein the LIDAR data comprises distances to reflective objects in an environment of the LIDAR sensor.

6. The LIDAR system of claim 1, further comprising: a bearing mount section configured to mount to the center portion of the rotational bearing.

7. The LIDAR system of claim 6, wherein the bearing mount section comprises a flange portion, wherein the flange portion is adapted to interface with the first waveguide section and the second waveguide section, and wherein the flange portion aligns the first waveguide section and the second waveguide section with the bearing waveguide.

8. The LIDAR system of claim 7, wherein the first waveguide section and the second waveguide section are in mechanical contact with the flange portion.

9. The LIDAR system of claim 1, wherein the first waveguide section and the second waveguide section are separated from the bearing waveguide by an air gap.

10. The LIDAR system of claim 1, wherein the first waveguide section, the second waveguide section, and the bearing waveguide are configured to propagate the electromagnetic signals at a frequency between 77 and 81 Gigahertz.

11. The LIDAR system of claim 1, wherein the bearing waveguide is configured to rotate with respect to the first waveguide section.

12. The LIDAR system of claim 1, wherein the bearing waveguide is configured to rotate with respect to the second waveguide section.

13. The LIDAR system of claim 1, wherein the first waveguide section, the second waveguide section, and the bearing waveguide are circular waveguides.

14. A vehicle comprising:

a computing device;

a light detection and ranging (LIDAR) system coupled to the vehicle, wherein the LIDAR system comprises:

22

a non-rotational unit having a first waveguide section, wherein the non-rotational unit is coupled to the vehicle at a fixed position;

a rotational unit having a second waveguide section; and

a rotary joint having a bearing waveguide located in a center portion of a rotational bearing of the rotary joint,

wherein the rotational bearing is configured to allow the rotational unit to rotate with respect to the non-rotational unit,

wherein the bearing waveguide is aligned with the first waveguide section and the second waveguide section such that electromagnetic signals are able to propagate between the rotational unit and the non-rotational unit, and

wherein the rotational unit includes a LIDAR sensor, and wherein the electromagnetic signals comprise LIDAR data from the LIDAR sensor.

15. The vehicle of claim 14, wherein the electromagnetic signals further comprise control signals for controlling the LIDAR sensor.

16. The vehicle of claim 14, wherein the LIDAR sensor comprises one or more laser sources and one or more detectors.

17. The vehicle of claim 16, wherein the LIDAR sensor further comprises a laser scanner.

18. A method comprising:

coupling electromagnetic signals between a non-rotational unit and a rotational unit via a rotary joint,

wherein the non-rotational unit includes a first waveguide section, the rotational unit includes a second waveguide section, and the rotary joint includes a bearing waveguide located in a center portion of a rotational bearing of the rotary joint,

wherein the rotational bearing is configured to allow the rotational unit to rotate with respect to the non-rotational unit and the non-rotational unit is coupled to a vehicle at a fixed position,

wherein the bearing waveguide is aligned with the first waveguide section and the second waveguide section such that electromagnetic signals are able to propagate between the rotational unit and the non-rotational unit, and

wherein the rotational unit includes a light detection and ranging (LIDAR) sensor, and wherein the electromagnetic signals comprise LIDAR data from the LIDAR sensor.

19. The method of claim 18, wherein the electromagnetic signals further comprise control signals for controlling the LIDAR sensor.

20. The method of claim 18, wherein the LIDAR data comprises distances to reflective objects in an environment of the LIDAR sensor.

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